Journal of English for Academic Purposes 73 (2025) 101466

Contents lists available at ScienceDirect

Journal of English for Academic Purposes

journal homepage: www.elsevier.com/locate/jeap

‘X can be classified into … ’: A local grammar of classification in academic discourse and its implications for EAP pedagogy

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ARTICLE INFO

Handling Editor: Dr Hilary Nesi

Keywords:
Local grammar
Classification
Academic discourse
EAP research

ABSTRACT

This study employs the local grammar approach to account for classification in academic writing and discusses the implications and applications for EAP writing pedagogy. Instances of classification were extracted from the Database of English for Academic Purposes (DEAP) Baby Corpus using classificatory phrases, nouns, and verbs (e.g., *fall into*, *type*, *classify*). Based on the retrieved instances, nine functional terms for classification (e.g., ‘Subclass’, ‘Superclass’, ‘Classifying’) were identified for performing local grammar analyses. The subsequent analyses produced 24 local grammar patterns of classification, each representing the lexical-semantic regularities of a group of instances in the corpus. A comparison between the local grammar patterns and previous patterns of classification indicates that the former can more accurately describe the linguistic regularities of academic discourse acts. Furthermore, local grammar patterns with different levels of complexity can be used to gradually improve students’ academic writing skills. This study also sheds new light on terminology identification for local grammar analyses of discourse acts that involve complex semantic relations, such as hyponymy and meronymy.

1. Introduction

Classifying items or concepts into categories is “a crucial skill in any kind of writing, and particularly academic writing” (Murray, 2012, p. 53). This skill of writing, as suggested by Murray and Hughes (2008), involves a crucial discourse act, i.e., classification, which serves to organize concepts by “classifying items into clearly identifiable groups” (p. 26). In this study, the term ‘discourse act’ refers to “communicative or rhetorical functions such as definition, exemplification, summarization, and so on” (Su et al., 2021, p. 21). The discourse act of classification can facilitate the understanding of the shared “qualities and features” of terminologies in academic writing, as well as the “differences between things” (Morley, 2023, p. 99).

To clarify the definition of classification, it is essential to distinguish it from two other discourse acts that are easily confused with it: division and definition. Classification is closely associated with hyponymy, a semantic relation that emphasizes inclusion, whereas division is more aligned with meronymy, which describes part-whole relationships (see Darian, 1997, p. 832). For example, the sentence *the cake is divided into five parts* illustrates division rather than classification, as it describes a part-whole relationship (i.e., the cake has five parts). The distinction between classification and definition can sometimes be blurred, as definition and classification can both be in the form of ‘X is a Y’ (Murphy & Koskela, 2010). However, as Cruse (2000) notes, ‘X is a Y’ serves more as a definition than a classification, as in *a horse is an animal*, whereas classification should contain a clear classificatory marker (e.g., *type*, *kind*, and *category*), characterized by the form ‘X is a type/kind/category of Y’ (e.g., *a horse is a type of animal*). This study will focus exclusively

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https://doi.org/10.1016/j.jeap.2024.101466
Received 6 May 2024; Received in revised form 2 December 2024; Accepted 5 December 2024
Available online 5 December 2024
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on classification instances that include classificatory markers. More specifically, following Darian's (1997) analysis, this study examines classifications that consist of at least three key meaning components: 1) an item being classified, 2) a classificatory marker (e.g., *fall into*, *type*, *classify*, etc.), and 3) a category that the classified item falls into.

A substantial body of research has contributed valuable insights into the lexicogrammatical patterns associated with classification in both EAP teaching and corpus linguistics. In the context of EAP teaching, studies have systematically highlighted the importance of teaching the phraseological patterns of classification to help students organize complex information (e.g., Faber, 2022; Gil Berrozpe, 2023; Murray, 2012). Relevant studies have also emphasized the use of the discourse act in developing analytical skills and critical thinking (e.g., Coffin et al., 2005; Flachmann et al., 2007; Kiniry & Rose, 1993). In corpus linguistics, research has successfully identified a variety of classificatory markers (e.g., Bielinskiene et al., 2012; Hearst, 1992; Pearson, 1998) and lexicogrammatical patterns of classification (e.g., Darian, 1997; Fang, 2021; León-Araúz et al., 2016). Despite the value of these studies, certain limitations can be identified across the literature. For example, the classification patterns identified by a considerable number of studies (e.g., Bielinskiene et al., 2012; Darian, 1997; Murray & Hughes, 2008) are overgeneralized or oversimplified. Meanwhile, a majority of recent studies (e.g., Fang, 2021; León-Araúz et al., 2016; León-Araúz et al., 2018) focused solely on lexicogrammatical features of classification, yet the semantic patterns of classification require more investigation. To address these limitations, a more robust linguistic analytical tool needs to be employed to reveal the "lexical-semantic relation" in classification instances (Bai et al., 2021, p. 2912). To be more specific, effectively performing this discourse act in academic writing requires careful use of lexical-semantic patterns to clarify certain key meaning components, such as what needs to be classified, what marks the act of classification, and what categories the classified items fall into.

To this end, I apply the local grammar approach in corpus linguistics pioneered by Hunston and Sinclair (2000) to investigate classification. Recent research has shown that local grammar, "a grammar of a discourse function" (Hunston & Su, 2019, p. 571), "not only allows us to explore discourse acts at the lexicogrammatical level, but also at the discourse-semantic level" (Su & Zhang, 2020, p. 4). In light of this, local grammar can be particularly suitable for the lexical-semantic investigation into classification. Darian (1997) observed that in academic discourse, "a frequent pattern is a definition, followed by a classification—at one or more levels—terminating in specific examples" (p. 819). To date, local grammars of definition and exemplification have been well developed (e.g., Barnbrook, 2002; Su & Zhang, 2020; Zhang & Su, 2022), but the crucial link that connects definition with exemplification, i.e., classification, remains an uncharted territory. In this paper, I aim to bridge this gap by investigating the discourse act of classification from a local grammar perspective and further discuss its implications for EAP pedagogy.

2. Review of research on classification

2.1. Classification in EAP teaching

Classification is "vital to language understanding and meaning representation" (Gil Berrozpe, 2023, p. 41) because it breaks down complex meanings into smaller, more manageable parts and organizes concepts systematically. This process is particularly valuable in academic discourse, which frequently involves explaining sophisticated terminologies (see Faber, 2022; Madsen & Thomsen, 2009; Nirenburg & Raskin, 2004). Since classification structures complex meanings, and academic writing often requires dealing with such complexities, this discourse act has long been a focus in EAP teaching (e.g., Hamp-Lyons & Heasley, 2006; Morley, 2023; Murray, 2012; Murray & Hughes, 2008). Most studies on classification in EAP settings focus on the phraseologies used to perform this discourse act. For instance, Hamp-Lyons and Heasley (2006) highlight three frequently used phraseologies for classification, i.e., *be types of*, *be classified into*, and *be classified according to*. These phraseologies are practiced in various exercises, including identifying classification in texts and converting visual representations of classification (such as classification trees) into written academic discourse. Similarly, Murray and Hughes (2008) provide six phraseologies associated with classification, such as *X can be classified into* and *X can be categorized/classified/grouped according to*. Murray (2012) further included ten phraseologies based on classificatory verbs in his EAP teaching practice of "handy language for classifying" (p. 53). More recently, in the well-known Academic Phrasebank led by Morley (2023), similar phraseologies are categorized into three groups: classifying a topic, specific classifications, and commenting on a classification system.

Classification, or categorization, is an essential cognitive ability (Lakoff, 1987; Taylor, 1995). This cognitive ability requires humans to critically assess and systematically organize concepts in both the physical and intellectual world (see Faber et al., 2014; Faber & López Rodríguez, 2012; Meteyard et al., 2012). For this reason, classification has often been applied to develop students' analytical skills and critical thinking in EAP teaching (e.g., Coffin et al., 2005; Flachmann et al., 2007; Kiniry & Rose, 1993; Rose, 2013). In Kiniry and Rose's (1993) EAP course design, for example, students are asked to reclassify the already established categories. The goal is to move students from a state of passive reception of knowledge to active participation in creating and understanding the categories. Similarly, in the work of Coffin et al. (2005), classification is presented as a method to dissect complex information, make informed judgments, and communicate more effectively. Flachmann et al. (2007) enhance students' analytical skills by assigning tasks that require them to classify terminologies, thereby helping them make connections and identify underlying principles in various disciplines. More recently, Rose (2013) integrated the discourse act of classification into developing students' analytical skills, using simple tasks like categorizing paintings without artist or period information to help students build foundational analytical abilities. After finishing these simple tasks, he introduced classification phraseologies and let students practice using them in complex EAP contexts.

2.2. Classification in corpus linguistics

A significant body of research in corpus linguistics has concentrated on the automatic or semi-automatic retrieval of classification expressions using phraseological markers (e.g., Bielinskiene et al., 2012; Hearst, 1992, 1998; Pearson, 1998). Hearst (1992) conducted one of the pioneering studies on classification, using the phrase 'is a' to automatically identify classification instances, demonstrating the potential of specific phraseological markers in capturing hyper-hyponymic word pairs. Pearson (1998) used a broader range of classificatory verbs, such as *classify*, *categorize*, and *group*, to identify classification instances. Building on these foundations, Bielinskiene et al. (2012) further extended the range of verbal markers to include verbs like *include* and *contain*, enhancing the comprehensiveness of classification retrieval processes. These studies collectively demonstrate the effectiveness of using a diverse set of classificatory markers to capture the complex linguistic regularities of classification.

A number of corpus studies have further identified the lexicogrammatical patterns of classification (Darian, 1997; Fang, 2021; León-Araúz et al., 2016; León-Araúz et al., 2018; Lewis, 2019). For example, Darian (1997) proposed six patterns of classification based on lexical markers such as *classify*, *categoris(z)e*, and *include*. However, his patterns tend to be either overly broad (e.g., 'Enumeration of Classes') or unnecessarily detailed (e.g., 'Superordinate + be + either + Hyponym + or + Hyponym'), resulting in a lack of granularity or redundancy. Meanwhile, a majority of recent studies (Fang, 2021; León-Araúz et al., 2016; León-Araúz et al., 2018) tend to neglect the fact that instances containing the same phraseologies may serve different functions, while instances with different phraseologies may fulfill the same function. For example, León-Araúz et al. (2016) distinguish between 'types|kinds| … of HYPERNYM are HYPONYM' and 'type|kind| … of HYPERNYM is HYPONYM'. Despite the lexical differences between the singular and plural forms of classificatory nouns (e.g., *type*, *kind*), the functions/meanings of these patterns do not seem to change much. This suggests that the two patterns should be merged into one. Similarly, Fang (2021) presented several phraseological patterns of classification like '[…] is classified into […] and […]' and '[…] is organized into: (1) […]; (2) […]; (3) […]'. However, these patterns can be consolidated into a single pattern due to their similar meaning sequences.

2.3. Research gap and the current study

The aforementioned studies have documented important phraseological or lexicogrammatical patterns of classification. However, there are three main limitations in the identified patterns: 1) Some patterns are either overgeneralized or oversimplified. 2) Most studies extract instances of classification using a very limited set of classificatory markers; few combine different phraseological, nominal, and verbal markers to form a list of classificatory markers that is as comprehensive as possible. This narrow focus can lead to an incomplete identification of classification patterns. 3) Previous research has predominantly emphasized the lexicogrammatical differences in the realizations of classification without considering the deeper semantic similarities, resulting in redundant patterns. This study aims to address these issues by using a more comprehensive list of classificatory markers to extract instances from the corpus and then mapping semantic roles onto the corresponding linguistic forms of classification. To achieve this objective, a robust analytical tool is warranted. One potential option could be the approach of local grammar in corpus linguistics, pioneered by Hunston and Sinclair (2000). I shall review relevant studies to illustrate the strengths of this analytical approach in the following section.

3. Local grammar approach to discourse acts in academic writing

Local grammar, an analytical tool in corpus linguistics, can describe the lexical-semantic realizations of language used for specific communicative purposes (Hunston & Su, 2019). This approach differs from other grammatical frameworks, e.g., pattern grammar, construction grammar, SFG, and TG grammar, which examine language as a whole. Instead, local grammar often focuses on particular speech or discourse acts, such as definition, evaluation, or exemplification, and aims to identify their unique lexical-semantic patterns in large corpora (Hunston, 2002). Local grammar analysis involves mapping transparent and context-specific functional terminologies onto formal elements. Table 1 shows an adaptation of Su and Zhang's (2020) local grammar analyses of exemplification, which maps terminologies like 'Exemplification' and 'Indicator' onto their structural elements, producing a set of local grammar patterns such as 'Exemplification + Hinge + Indicator + Exemplified' to describe the language of exemplification from a lexical-semantic perspective.

Local grammar labels forms based on their functions in specific contexts, whereas other grammatical frameworks often use a fixed set of terms for all types of speech/discourse acts. For example, in the pattern 'V n to-inf' identified by pattern grammar (Francis, 1993; Hunston, 2015; Hunston & Francis, 2000), the verbs are classified into 12 groups, such as 'verbal processes' (e.g., *ask*, *tell*), 'causation' (e.g., *cause*, *compel*), and 'helping' (e.g., *aid*, *help*) (Francis et al., 1996). However, the meaning tags of the words are context-free and can be inaccurate. To illustrate, consider the following two sentences: (1) *The boss told her to finish the report by noon.* (2) *Her friend told her to rest more after her illness.* The verb *told* fulfills the function of 'instruction' in the first sentence but 'advice' in the second. However, pattern grammar fails to capture this difference in function, putting the two different *told* into the same 'verbal processes'

Table 1
Local grammar analyses of exemplification adapted from Su and Zhang (2020).

Exemplification The incident	**Hinge** was	**Indicator** an example of	**Exemplified** the financial crisis.
Indicator An instance of	**Exemplified** this concept	**Hinge** is	**Exemplification** personification.

category. Local grammar, on the other hand, "assigns category labels that are far more transparent and trustworthy than the highly general ones" (Hunston & Sinclair, 2000, p. 178). This is because local grammar "is always a grammar of a discourse function" and is "closely related to performative speech acts" (Hunston & Su, 2019, p. 571). From a local grammar perspective, the different functions of *told* in the two sentences above can be described by using the terminology 'instructing' for the first sentence's *told* and 'advising' for the second.

Local grammar has been applied to account for various discourse acts in EAP settings, such as cause and effect (Allen, 2005), definition (Zhang & Su, 2022), exemplification (Su & Zhang, 2020), and graphic data commentary (Zhang et al., 2024), among others. Allen's (2005) study made one of the pioneering contributions to the application of local grammar in the academic context. Based on a two-million-word corpus of biomedical scientific texts, Allen developed a local grammar of cause-and-effect in academic discourse and tested its potential applications in natural language processing. Allen's (2005) study also points out the effectiveness of local grammar in parsing academic texts, aiming to enhance automated analysis and information retrieval in scientific writing. Zhang and Su (2020) identified 17 distinct patterns of exemplification in a corpus of linguistics research articles. Notably, their study also explored the value of local grammar in EAP teaching, stating that local grammar can provide an inventory of language resources for academic writers to effectively perform discourse acts. For the discourse act of definition in linguistic research articles, Zhang and Su (2022) identified 36 lexical-semantic patterns from a local grammar perspective. They further explored the pedagogical value of local grammar, suggesting that an entire textbook based on local grammar can be developed to facilitate teaching discourse acts in EAP settings, with each chapter dedicated to a particular discourse act. Most recently, drawing on a corpus of economics research articles, Zhang et al. (2024) identified 17 frequently occurring local grammar patterns for commenting on graphical data. These patterns systematically represent the ways economists report and interpret graphic data. Zhang et al.'s (2024) research also proposed four specific pedagogical steps to teach local grammar patterns in EAP writing courses.

A major advantage of local grammar is that it can "accurately measure the lexicogrammatical expressions of a specific discourse act" (Su et al., 2021, p. 123). Instead of only counting the number of lexical markers of a discourse act, local grammar offers a more reliable way to measure the realizations of a discourse act (Su & Lu, 2022). This is because instances containing the same marker may have different meanings or functions, and instances containing different markers may have the same meanings or functions. I now employ a local grammar of exemplification (Su & Zhang, 2020) to demonstrate the strengths of this analytical tool over traditional lexicogrammatical analyses. Tables 2 and 3 present two local grammar analyses of exemplification based on Su and Zhang (2020). As shown in Table 2, *as illustrated* and *for example* are both labeled as 'Indicator' because they both realize the function of "signaling the act of exemplifying" (Su & Zhang, 2020, p. 5), whereas these two phrases are often analyzed separately from a lexicogrammatical perspective due to their different forms. It can be seen that a local grammar approach can emphasize the shared functions between expressions like *as illustrated* and *for example*, despite their formal differences.

On the other hand, phrases with the same form (e.g., *an instance of*) may be analyzed as the same realization of a discourse act from a traditional lexicogrammatical perspective, but serve very different functions. For example, in Table 3, the phrase *an instance of* is in a clause labeled as 'Exemplified', which fulfills the function of "the proposition or superordinate category that is being illustrated, clarified, or elaborated" (Su & Zhang, 2020, p. 5). However, in the second instance, the same phrase *an instance of* is assigned a different terminology 'Indicator', as it realizes a different function of "signaling the act of exemplifying" (Su & Zhang, 2020, p. 5).

Darian (1997) observes that academic discourse often follows a sequence of definition, classification, and exemplification. It can be seen from the abovementioned studies that local grammar has already been applied to the studies on definition and exemplification, but the crucial link between them, i.e., classification, has not been analyzed under the lens of local grammar. In this paper, I aim to fill this gap by analyzing classification with a local grammar approach and further discussing its implications for EAP teaching.

4. Corpus and research method

4.1. Corpus

To systematically analyze the instances of classification in academic discourse, I drew upon data from the Database of English for Academic Purposes (DEAP) Baby Corpus (Xu & Sun, 2021), which is a sub-corpus of the DEAP Corpus (https://corpus.bfsu.edu.cn/info/1082/1561.htm). The DEAP Corpus, built by the Corpus Research Group at Beijing Foreign Studies University in mainland China, comprises English research articles from 25 disciplines published in international journals of high impact factors between 2011 and 2021. However, the whole corpus totals 122,798,042 tokens, which is too large for the current study to handle. This is because after automatically extracting instances using lexical markers, each instance must be manually analyzed into a local grammar pattern. This requires the size of the corpus to be manageable. I therefore opted for a sub-corpus of DEAP, i.e., the DEAP Baby Corpus. This is a balanced multi-discipline English for Academic Purposes corpus developed by members from the same team that built the DEAP Corpus. The DEAP Baby Corpus is derived from the 125-million-word DEAP Corpus through resampling, with 50,000 words sampled from each of the 25 disciplines. It comprises 1,424,581 tokens and 82,273 types. More details about this corpus are available at http://

Table 2
A local grammar analysis of exemplification (1) adapted from Su and Zhang (2020).

Exemplified	Indicator	Exemplification
Indefinite nouns do occur with case particles,	as illustrated	in examples (1), (2) and (3).
These particles include,	for example,	the stance students convey towards their artefacts.

Table 3
A local grammar analysis of exemplification (2) (adapted from Su & Zhang, 2020).

Researcher	**Exemplifying**	**Exemplified**	
We	exemplify	how an instance of these markers fits into the context.	
Exemplification	**Hinge**	**Indicator**	**Exemplified**
This sentence	is	an instance of	personification.

corpus.bfsu.edu.cn/info/1082/1813.htm.

The DEAP Baby Corpus is selected for several key reasons. Firstly, the texts within the corpus are all published research articles that have undergone rigorous peer review, ensuring that the research results of this study can accurately represent expert-level academic writing and can be effectively utilized for teaching the discourse act of classification. Secondly, the corpus contains expert writing across a broad spectrum of disciplines, enabling a comprehensive analysis across diverse academic fields. Thirdly, as mentioned above, the size of the corpus is manageable. Finally, this corpus is available for free download (http://corpus.bfsu.edu.cn/DEAPBabyV1.zip), thereby enhancing the accessibility and reliability of the study's data retrieval and fostering replicability and reproducibility of this research.

4.2. Working procedure

4.2.1. Data retrieval

Previous research (e.g., Darian, 1997; Hearst, 1992, 1998; Bielinskiene et al., 2012; León-Araúz et al., 2016) shows that the performance of classification often involves certain phrasal markers (e.g., *fall into*, *fall within*, and *belong to*), nominal markers (e.g., *type*, *category*, and *kind*) and verbal markers (e.g., *classify*, *categorize*, and *include*). Drawing on these studies, I employed a wide range of language markers to extract classification instances in the DEAP Baby Corpus. These markers were categorized into three main groups: classificatory phrases, classificatory nouns, and classificatory verbs. Table 4 presents the markers in each category along with their raw and normalized frequencies in the corpus (per million words). Note that the frequencies of markers which contain verbs count not only the bare forms of the verbs, but also their preterite, past participle, present participle, and other morphological variants. For nominal markers, both singular and plural forms are included.

These predetermined markers served as a foundation for identifying classification instances within the corpus. Admittedly, relying

Table 4
Markers of classification and their frequencies in the DEAP Baby Corpus.

Marker type	Marker	Raw frequency	Normalized frequency
Classificatory phrase	belong to	90	63
	fall into	18	13
	see as	75	53
	regard as	53	37
	consider as	39	27
	identify as	43	30
	perceive as	28	20
	recognize as	15	11
	label as	10	7
	fall under	5	4
	fall within	3	2
Classificatory noun	type	907	637
	category	310	218
	kind	217	152
	variety	164	115
	classification	157	110
	branch	67	47
	sort	65	46
	subgroup	52	37
	subtype	10	7
	subfamily	4	3
	subcategory	5	4
	subclass	2	1
	subspecies	1	1
Classificatory verb	include	1366	959
	contain	467	328
	classify	106	74
	organis(z)e	87	61
	group	55	39
	categoris(z)e	39	27
	subclassify	0	0
	subcategoris(z)e	0	0
Total		4460	3131

on these pre-determined markers cannot exhaustively capture all instances within the corpus since classification can be realized without these markers. However, as discussed in Section 2, prior studies have demonstrated that instances of classification typically co-occur with specific lexical items (e.g., Bielinskiene et al., 2012; Hearst, 1992, 1998; Pearson, 1998). Moreover, classification instances that contain classificatory phrases, nouns, and verbs, as noted by Darian (1997) and León-Araúz et al. (2016), can be the most representative ones and thus merit investigation.

Then, I used the BFSU CQPweb (http://114.251.154.212/cqp/deapbaby/) to search for different forms of the abovementioned markers to retrieve instances of classification. Each automatically retrieved instance was manually examined to eliminate those that do not perform the discourse act of classification. The screening process was guided by Darian's (1997) definition of a typical classification, which must include at least the following three semantic elements: 1) an item being classified, 2) a classificatory marker (e.g., *fall into*, *type*, *classify*, etc.), and 3) a category that the classified item falls into. Based on this criterion, 1396 instances were identified and further put into local grammar analyses. The remaining 3064 instances were excluded, primarily because they fail to encompass all three meaning components. For example, in the instance *It is therefore possible to classify non-invasive cancers*, the verb *classify* is a classificatory marker and the noun phrase *non-invasive cancers* represents the item being classified. However, the category into which the item is classified is not specified in the instance, which was therefore deleted and would not be put into further local grammar analysis.

4.2.2. Terminology identification and local grammar analysis

After carefully reading the retrieved instances from the corpus, I summarized the semantic elements in the instances into functional terminologies based on Su's (2020) guiding principles of terminology identification. For illustration, consider the following two instances in the corpus.

(1) **The spectral data** are classified as **true data.**
(2) **Cohesion** is a kind of **coping resource.**

In Examples 1 and 2, the first noun phrases (i.e., *the spectral data*, *cohesion*) are sub-categories or specific instances within broader categories and were therefore termed 'Subclass'. On the other hand, *true data* and *coping resource* are superordinate or general categories. I thus further abstracted them into 'Superclass'. In this manner, nine functional terms were identified from the corpus instances, as shown in Table 5. While most terminologies are explicit and self-explanatory, the 'Hinge' concept needs further clarification. Following Hunston and Su (2019, p. 582), 'Hinge' in this study is defined as the element that connects different functional terms. However, this definition alone can be vague in the practice of local grammar analyses when determining whether certain phrases should be annotated as 'Hinge'. To address this, I carefully examined previous studies that adopt Hunston and Su's (2019) definition of 'Hinge', and identified three specific groups of words commonly annotated as 'Hinge': 1) copular or linking verbs (e.g., *be*, *am*, *is*, *are*, *was*, *were*) (e.g., Pontrandolfo & Goźdź-Roszkowski, 2014, p. 76), 2) modal verbs (*can*, *could*, *will*, *would*, *shall*, *should*, *may*, *might*) (e.g., Su, 2017, p. 77; Su et al., 2022, pp. 7, 9), and 3) auxiliaries (e.g., *do/does/did*, *have/has/had*) (e.g., Hunston & Su, 2019, p. 582). In this study, words from these three groups were generally analyzed as 'Hinge'. It should be noted that to avoid overcomplicating the analysis, the current study would not analyze prepositions independently as 'Hinge' (see Hunston & Su, 2019).

The last step was to allocate the local grammar terminologies in Table 5 to each structural element of the retrieved instances. As Table 6 illustrates, Example 2 was analyzed as follows: *Cohesion* is a sub-category to be classified, and was therefore labeled as 'Subclass'. The copular 'be' verb (i.e., *is*) was annotated as 'Hinge'. The 'of' phrase containing a nominal marker of classification (i.e., *a kind of*) was analyzed as 'Indicator'. The last part (i.e., *coping resource*) represents the superordinate category that *cohesion* falls into and was therefore labeled 'Superclass'. This analysis resulted in the local grammar pattern of 'Subclass + Hinge + Indicator + Superclass', representing the semantic regularity of a group of instances. In this way, I analyzed all the retrieved instances into a set of local grammar patterns, which can systematically document the lexical-semantic regularities of classification in the DEAP Baby Corpus.

Table 5
Local grammar terminologies of classification.

Terminology	Definition	Example
Subclass	The subordinate category within a superordinate category	***Partner proteins*** *belong to the ETS protein family.*
Superclass	The superordinate category that contains various subordinate categories	*The spectral data are classified as* ***true data****.*
Target	The entity being classified, without a clear distinction as to whether it is a subordinate or superordinate category	*Stern classifies* ***the environmentally significant behavior*** *as private-sphere environmentalism.*
Category	The element that indicates what an entity falls into, often specifying the number of categories an entity can be classified into	*Land covers are classified into* ***44 classes****.*
Indicator	The 'of' phrase containing nominal markers of classification, such as *a type of*, *the kind of*, and *categories of*	*Computer programming is* ***a type of*** *story sequencing.*
Classifying	The act of making classification, typically realized by classificatory verbs	*We* ***classify*** *microfluidic experimentation into three generic stages.*
Categorizer	The performer of classification	***We*** *classify microfluidic experimentation into three generic stages.*
Basis	The standard, criterion, or principle that forms the basis of classification	*A policy π can be classified according to* ***Sπ and Rπ****.*
Hinge	The connecting element between terminologies	*PRRs* ***can be*** *classified into five major groups (CLRs, TLRs, RLRs, NLRs and ALRs).*

Table 6
Local grammar analysis of Example 2.

Subclass	Hinge	Indicator	Superclass
Cohesion	is	a kind of	coping resource.

5. Results

Following the working procedures described in Section 4, I identified 24 local grammar patterns of classification. These patterns can be categorized into three main groups based on the directionality of classification. In Group 1, the patterns move from specific ('Subclass') to general ('Superclass'), while in Group 2, they move from general ('Superclass') to specific ('Subclass'). Group 3, however, demonstrates no clear classification direction or superclass-subclass relation. To present the results more clearly, I further divided each main group into three subgroups. Below are the descriptions of the 24 local grammar patterns.

Group 1, characterized by specific-to-general classification, includes Patterns 1–8. Patterns 1–4 (see Tables 7–10) belong to the first subgroup of Group 1, which are among the most concise patterns in terms of the number of meaning components. They only retain the most fundamental three or four semantic elements to perform the discourse act of classification. Pattern 1 is characterized by the use of phrases such as *belong to* and *fall into* as 'Classifying'. Pattern 2 is similar to Pattern 1, with the only difference lying in the addition of 'Indicator' before 'Superclass', which clearly marks the performance of classification. Pattern 3 is slightly different from the first two patterns. It uses a 'be' verb as 'Hinge', and often a classificatory noun at the end of 'Superclass'. Pattern 4 is based on Pattern 3. Similarly to the way Pattern 2 is derived from Pattern 1, Pattern 4 adds 'Indicator' before 'Superclass'. These patterns serve as straightforward ways to perform classification, enabling readers to quickly comprehend the underlying hyper-hyponymic or inclusion relationships of concepts.

Patterns 5 and 6 fall into the second subgroup under Group 1. As shown in Tables 11 and 12, these two patterns are in the passive voice, and 'Superclass' in these patterns usually begins with the preposition *as*. These patterns are especially useful when the act of classification is more important than the categorizer or when the categorizer is assumed to be implicit, as is often the case in scientific reporting. Pattern 6 differs from Pattern 5 in providing a classification basis at the end, which provides additional information about how or why an entity is classified. 'Basis' is often initiated by compound prepositions such as *according to*, *based on/upon*, and *depending on/upon*.

Patterns 7 and 8 (Tables 13 and 14), unlike the previous two patterns in the passive voice (Patterns 5 and 6), employ the active voice to explicitly involve the agent or basis in the classification. In Pattern 7, 'Categorizer' at the beginning highlights the agent's influence on the classification process. This pattern is particularly useful when the categorizer's identity is relevant, and the act of classification needs to be clearly attributed. By highlighting the categorizer, Pattern 7 provides transparency and personal accountability, making it ideal for contexts where the authority and responsibility of the categorizer are important. Pattern 8 differs from Pattern 7 in the first element. Pattern 8 begins with 'Basis' rather than 'Categorizer', which actively grounds the classification in a specific rationale or methodology.

The second main group, encompassing Patterns 9–18, demonstrates a general-to-specific classificatory direction. Within this group, Patterns 9–11 (Tables 15–17) form the first subgroup. The 'Classifying' element in Pattern 9 is often realized by *include*. Pattern 10 replaces 'Classifying' in Pattern 9 with 'Hinge', often realized by a 'be' verb. Another notable feature of Pattern 10 is that the 'Superclass' in this pattern usually ends with classificatory nouns like *categories* or *types*. Pattern 11 builds upon Pattern 10 by incorporating 'Indicator' before 'Superclass', transparently marking the performance of classification. From Patterns 2, 4, and 11, it can be seen that 'Indicator' is typically positioned next to 'Superclass' rather than 'Subclass'.

The second subgroup under Group 2 includes Patterns 12–14. As shown in Tables 18–20, these three patterns resemble Patterns 5 and 6, but demonstrate an opposite direction of classification. 'Subclass' in these patterns often begins with the preposition *into*. Similarly to the way Pattern 6 is derived from Pattern 5, Pattern 13 builds on Pattern 12 by incorporating an additional semantic element ('Basis') to provide a rationale for the classification. Pattern 14 differs from the other two patterns in this subgroup. Instead of transitioning directly from 'Classifying' to 'Subclass', it adds a 'Category' element in between to summarize how many categories a superordinate category ('Superclass') can be classified into. Following this, a colon is often used to introduce the specific subcategories ('Subclass'), or alternatively, parentheses may be used to enclose the subcategories.

Patterns 15–18 (Tables 21–24) belong to the third subgroup of Group 2. These patterns, although similar to those in the third subgroup of Group 1 in the way they emphasize either the performer or the classification basis, distinguish themselves by moving from general to specific, with the preposition *into* often used at the beginning of 'Subclass'. Pattern 17, similar to Pattern 14, includes a 'Category' element before 'Subclass' to summarize the number of categories into which a 'Superclass' is classified.

The patterns in Group 3 show no clear classification direction, using two terms (i.e., 'Target' and 'Category') that involve no super-sub relations. Pattern 19 alone forms the first subgroup under Group 3. This pattern resembles Patterns 1 and 9, often using *fall into*,

Table 7
Pattern 1 'Subclass + Classifying + Superclass'.

Subclass	Classifying	Superclass
Partner proteins	belong to	the ETS protein family.
The tests used to deform graphene	fall into	the tension test category.

Table 8
Pattern 2 'Subclass + Classifying + Indicator + Superclass'.

Subclass	Classifying	Indicator	Superclass
Epigenetic modifications	fall into	the class of	molecular mechanisms.
Immunotherapeutic interventions	belong to	the category of	biological treatments.

Table 9
Pattern 3 'Subclass + Hinge + Superclass'.

Subclass	Hinge	Superclass
Cylinder arrangements	are	V type.
The signal noise	was	Gaussian type.

Table 10
Pattern 4 'Subclass + Hinge + Indicator + Superclass'.

Subclass	Hinge	Indicator	Superclass
Computer programming	is	a type of	story sequencing.
His playing	is	a kind of	spectatorial identification.

Table 11
Pattern 5 'Subclass + Hinge + Classifying + Superclass'.

Subclass	Hinge	Classifying	Superclass
The spectral data	are	classified	as true data.
Australia	is usually	classified	as a "liberal" welfare state.

Table 12
Pattern 6 'Subclass + Hinge + Classifying + Superclass + Basis'.

Subclass	Hinge	Classifying	Superclass	Basis
A policy π	can also be	classified	as a deterministic policy	according to Sπ and Rπ.
Neon tetras	are	categorized	as characin fish	depending on their morphological features and the presence of an adipose fin.

Table 13
Pattern 7 'Categorizer + Classifying + Subclass + Superclass'.

Categorizer	Classifying	Subclass	Superclass
We	categorize	episodic memory	as declarative memory.
Environmental economists	often classify	carbon pricing mechanisms, such as carbon taxes and cap-and-trade systems,	as policy tools designed to internalize the environmental costs of greenhouse gas emissions.

Table 14
Pattern 8 'Basis + Classifying + Subclass + Superclass'.

Basis	Classifying	Subclass	Superclass
Plate tectonics theory	classified	the Mid-Atlantic Ridge	as a divergent boundary.
This electrical engineering model	classifies	capacitors	as energy storage devices.

Table 15
Pattern 9 'Superclass + Classifying + Subclass'.

Superclass	Classifying	Subclass
The manifestations of the disease	include	a swollen mouth, viscera ejection, skin ulceration and massive mortality.
Other software applications	include	Autodesk Green Building Studio, Integrated Environmental Solutions – Virtual Environment and Bentley AECOSim.

Table 16
Pattern 10 'Superclass + Hinge + Subclass'.

Superclass	Hinge	Subclass
Answer categories	were:	agree (0), neither agree nor disagree (1), or disagree (2).
Land cover types	are	(i) dry-field agriculture, (ii) dry-field agriculture mixed with grass (grassland that is suspected to have sporadic agricultural activities, e.g. shifting cultivation), (iii) dry-field shrub and grass and (iv) open land.

Table 17
Pattern 11 'Indicator + Superclass + Hinge + Subclass'.

Indicator	Superclass	Hinge	Subclass
Another type of	dominant ambient noise	is	thermal noise.
A new type of	fuel	is	hydrogenated catalytic biodiesel.

Table 18
Pattern 12 'Superclass + Hinge + Classifying + Subclass'.

Superclass	Hinge	Classifying	Subclass
Conflict resolution methods	are often	classified	into diplomacy, sanctions, and military interventions.
Climate change agreements	are generally	organized	into binding treaties, non-binding accords, and multilateral partnerships.

Table 19
Pattern 13 'Superclass + Hinge + Classifying + Subclass + Basis'.

Superclass	Hinge	Classifying	Subclass	Basis
The MP particles	were	classified	into "fiber", "shaft", "film", "flake" and "sphere"	according to their types.
Side branch resonators	can be	classified	into low-pass, high-pass, or band-stop filters,	depending on their impedance characteristics.

Table 20
Pattern 14 'Superclass + Hinge + Classifying + Category + Subclass'.

Superclass	Hinge	Classifying	Category	Subclass
Transfer learning	can be	categorized	into three categories:	(i) unsupervised transfer learning, (ii) transductive transfer learning and (iii) inductive transfer learning.
PRRs classified	can be	classified	into five major groups	(CLRs, TLRs, RLRs, NLRs and ALRs).

Table 21
Pattern 15 'Categorizer + Classifying + Superclass + Subclass'.

Categorizer	Classifying	Superclass	Subclass
We	have categorized	the risks	into operational, financial, and strategic risks.
We	categorized	the dataset	into training, validation, and test sets.

Table 22
Pattern 16 'Categorizer + Classifying + Superclass + Subclass + Basis'.

Categorizer	Classifying	Superclass	Subclass	Basis
We	can group	these behaviors	into proactive and reactive responses,	depending on the stimulus.
We	classify	the memory used in computing	into volatile and non-volatile memories,	based on their retention characteristics.

Table 23
Pattern 17 'Categorizer + Classifying + Superclass + Category + Subclass'.

Categorizer	Classifying	Superclass	Category	Subclass
We	classify	channels on chip	into four categories:	azimuthal channel, radial channel, left slant channel and right slant channel.
We	categorize	the consumer behaviors	into three categories:	pre-purchase, purchase, and post-purchase behaviors.

Table 24
Pattern 18 'Basis + Classifying + Superclass + Hinge + Subclass'.

Basis	Classifying	Superclass	Hinge	Subclass
This three-way categorization	classifies	judgments	into	'obvious' (Marantz's first two classes) and 'nonobvious' judgments (the third class).
The criteria for land use	organize	urban areas	into	residential, commercial, industrial, and mixed-use zones.

belong to or *include* to realize 'Classifying' (see Table 25). Patterns 20 and 21 (Tables 26 and 27) belong to the second subgroup under Group 3 and are characterized by their use of passive voice, like Patterns 5, 6, and 12–14. The third subgroup of Group 3 comprises Patterns 22–24 (Tables 28–30), which are characterized by using the active voice to emphasize 'Categorizer' or 'Basis', as in Patterns 7, 8, and 15–18.

6. Discussion

To facilitate further discussion of the 24 local grammar patterns, I collected all of them in Table 31. Also provided in the table are the classification directions and raw and normalized frequencies (per million words) of the 24 patterns.

Darian (1997) proposed six patterns of classification, some of which correspond directly to the local grammar patterns identified in this study. For example, his pattern 'Superordinate + be + classified into + Hyponym' aligns with Pattern 12 ('Superclass + Hinge + Classifying + Subclass'). However, the current local grammar study also differs from Darian's (1997) work, remedying some weaknesses of his study. The local grammar patterns in this study offer a finer-grained description by dividing overgeneralized patterns into more specific ones. For instance, Darian's (1997) pattern 'Hyponym + be + Superordinate' was specified into Pattern 3 'Subclass + Hinge + Superclass' and Pattern 4 'Subclass + Hinge + Indicator + Superclass', and his pattern 'Superordinate + be + Hyponym' was divided into Pattern 10 'Superclass + Hinge + Subclass' and Pattern 11 'Indicator + Superclass + Hinge + Subclass'. Meanwhile, some patterns in Darian's (1997) work (e.g., 'Superordinate + be + Hyponym + or + Hyponym') may contain some redundant elements. The conjunction *or* connects two coordinate elements that realize similar functions and therefore should be combined into a single element. For example, for the instance *answer categories were: agree (0), neither agree nor disagree (1), or disagree (2)*, the present study analyzed the part after the colon as a single meaning unit 'Subclass' rather than 'Subclass + Subclass + or + Subclass'. Also, Darian (1997) overlooked some important meaning units such as 'Categorizer' and 'Basis', while this study has considered them both, producing local grammar patterns such as Patterns 6–8, 13, 15–18, and 21–24.

This study also furthers the work of Bielinskiene et al. (2012), León-Araúz et al. (2016), Fang (2021), and Morley (2023). For instance, León-Araúz et al. (2016) identified 18 phraseological patterns of classification. However, many of these patterns differ only lexicogrammatically while realizing similar functions. For example, León-Araúz et al. (2016) distinguish between two patterns termed 'types|kinds| … of HYPERNYM are HYPONYM' and 'type|kind| … of HYPERNYM is HYPONYM'. However, these two patterns are similar from a lexical-semantic perspective. Therefore, this study consolidates them into the same local grammar pattern, 'Indicator + Superclass + Hinge + Subclass'. Likewise, many classification patterns given by Fang (2021) and Morley (2023) are somewhat scattered and lack systematic organization. For instance, Fang (2021) analyzed '[…] can be classified into […], […] and […]' and '[…] are organized into: (1) […]; (2) […]; (3) […]' as two separate patterns. However, the current local grammar study combined these similar patterns into Pattern 12 'Superclass + Hinge + Classifying + Subclass'.

Moreover, this study can provide further evidence for the value of local grammar in discourse act analysis. The advantages of local grammar in analyzing academic discourse acts can be fourfold. (1) Quantifiability. In this study, local grammar has described the discourse act of classification in the DEAP Baby Corpus into 24 local grammar patterns, which match 1396 instances (980 occurrences per million words). The quantitative information of each local grammar pattern is valuable for EAP teaching, as they can guide curriculum design to focus on the most frequent lexical-semantic patterns of classification and help students meet the most common communicative needs in a short period. (2) Precision. The local grammar analyses in this study show how each syntactic element of classification contributes to meaning-making, providing a precise description of language used for classification, with the functions of each pattern clearly specified (see also Hunston & Sinclair, 2000; Su & Wei, 2018). (3) Manageability. As discussed in this study, while the surface structures (specific instances) of classification may demonstrate considerable variety (up to 1396 different instances of classification), the core semantic elements and communicative functions are consistent (nine meaning units and 24 patterns). These lexical-semantic regularities documented by local grammar, as also suggested by Su and Zhang (2020), make the complex analyses and interpretation of discourse acts more manageable. (4) Enhancement of language acceptability and richness. The most frequent 4 patterns, i.e., Patterns 4, 9, and 11 have a total frequency of 862, accounting for 62% of all the instances. Increasing these highly frequent patterns in academic writing can "ensure a quicker recognition and comprehension" (Zhang et al., 2024, p. 78). On the other hand, less common patterns offer a variety of linguistic choices that can be used to diversify expressions and enhance language richness. The balance between the acceptability of language for ease of comprehension and the variety of language for language richness is an important use of the research results of local grammar studies of discourse acts.

In local grammar analyses of discourse acts, targets being acted upon and specific meanings expressed are typically labeled with terms like 'Target' and 'Evaluation' (Hunston & Su, 2019), 'Exemplified' and 'Exemplification' (Su & Zhang, 2020), and 'Definiendum' and 'Definiens' (Zhang & Su, 2022). However, for complex discourse acts involving intricate semantic relationships, such as classification—which relates to hyponymy or inclusion relations—relying solely on the terms 'Target' and 'Category' may prove insufficient. These terms can document classification patterns that lack superordinate-subordinate relations but fail to distinguish between the two opposite directions of classification. For instance, classifications that move from general to specific, such as *conflict resolution methods*

Table 25
Pattern 19 'Target + Classifying + Category'.

Target	Classifying	Category
Antecedents	fall into	two categories.
Channels on a microchip	include	four categories.

Table 26
Pattern 20 'Target + Hinge + Classifying + Category'.

Target	Hinge	Classifying	Category
Land covers	are	classified	into 44 classes.
The chemical compounds of the dataset	were	classified	into four categories.

Table 27
Pattern 21 'Target + Hinge + Classifying + Category + Basis'.

Target	Hinge	Classifying	Category	Basis
Nine stations	are	classed	into four categories	based on the lines of all stations.
Soils	can be	categorized	into 12 types	in terms of the fractions of the clay, sand, and silt.

Table 28
Pattern 22 'Categorizer + Classifying + Target + Category'.

Categorizer	Classifying	Target	Category
Stern	classifies	the environmentally significant behavior	as private-sphere environmentalism.
We	have grouped	countries	into five categories in Finland and six in Denmark and Sweden.

Table 29
Pattern 23 'Categorizer + Classifying + Target + Category + Basis'.

Categorizer	Classifying	Target	Category	Basis
We	sort	corporate bonds	into six groups	based on the average monthly trading volume in 2003.
Researchers	have classified	the study's participants	into five age groups	based on their respective age ranges.

Table 30
Pattern 24 'Basis + Classifying + Target + Category'.

Basis	Classifying	Target	Category
This standard of data organization	categorizes	entries	into three hierarchical groups.
The Bayesian inference model	classifies	protein structures	into 4 distinct folding patterns.

are often classified into diplomacy, sanctions, and military interventions, differ in classification direction from those that move from specific to general, as in *the spectral data are classified as true data.* Relying solely on traditional terms like 'Target' and 'Category' fails to account for this crucial difference in directionality, treating both instances similarly as 'Target + Hinge + Classifying + Category'. To overcome this limitation, this study introduces the terms 'Superclass' and 'Subclass', in addition to 'Target' and 'Category', to more accurately reflect the inherent directionality of classification processes. This refinement enhances the clarity and precision in documenting the regularities in meaning organization, thereby improving the theoretical and analytical rigor of local grammar studies. Furthermore, this method of terminology identification can shed light on future local grammar research on other discourse acts that involve complex semantic relationships, such as division (meronymy or part-whole relations), as well as comparison and contrast (synonymy and antonymy), among others.

7. The implications of the local grammar of classification for EAP pedagogy

In this section, I shall explore how the findings of this study can be applied to the design of English academic writing courses and inform the development of EAP teaching materials. Widdowson and Allen (1985) suggest that language learners need to know "how sentences are used in the performance of different communicative acts" (p. 74). For effective instruction of a communicative or discourse act, it is important to clarify both its linguistic forms and communicative functions (Pearson, 1998; Flowdew, 2017). Hence, in teaching the discourse act of classification, instructors should underscore not only "those classifying words—nominal and verbal—that help students recognize and understand the discourse pattern", but also how to use those words and linguistic patterns in the right context by understanding their communicative functions with "discourse-length material" (Darian, 1997, pp. 836–837). In this

Table 31
An overview of the local grammar patterns of classification in the DEAP Baby Corpus.

Directionality	Pattern	Frequency (Normed)
From specific ('Subclass') to general ('Superclass')	PA 1[a] Subclass + Classifying + Superclass	58 (41)
	PA 2 Subclass + Classifying + Indicator + Superclass	17 (12)
	PA 3 Subclass + Hinge + Superclass	44 (31)
	PA 4 Subclass + Hinge + Indicator + Superclass	128 (90)
	PA 5 Subclass + Hinge + Classifying + Superclass	54 (38)
	PA 6 Subclass + Hinge + Classifying + Superclass + Basis	18 (13)
	PA 7 Categorizer + Classifying + Subclass + Superclass	5 (4)
	PA 8 Basis + Classifying + Subclass + Superclass	5 (4)
From general ('Superclass') to specific ('Subclass')	PA 9 Superclass + Classifying + Subclass	509 (357)
	PA 10 Superclass + Hinge + Subclass	49 (34)
	PA 11 Indicator + Superclass + Hinge + Subclass	225 (158)
	PA 12 Superclass + Hinge + Classifying + Subclass	59 (41)
	PA 13 Superclass + Hinge + Classifying + Subclass + Basis	7 (5)
	PA 14 Superclass + Hinge + Classifying + Category + Subclass	10 (7)
	PA 15 Categorizer + Classifying + Superclass + Subclass	16 (11)
	PA 16 Categorizer + Classifying + Superclass + Subclass + Basis	6 (4)
	PA 17 Categorizer + Classifying + Superclass + Category + Subclass	11 (8)
	PA 18 Basis + Classifying + Superclass + Subclass	18 (13)
No super-sub relation	PA 19 Target + Classifying + Category	46 (32)
	PA 20 Target + Hinge + Classifying + Category	75 (53)
	PA 21 Target + Hinge + Classifying + Category + Basis	7 (5)
	PA 22 Categorizer + Classifying + Target + Category	23 (16)
	PA 23 Categorizer + Classifying + Target + Category + Basis	3 (2)
	PA 24 Basis + Classifying + Target + Category	3 (2)
Total		1396 (980)

[a] PA = Pattern.

sense, the 24 local grammar patterns given in this study, which have provided both lexicogrammatical and discourse-semantic descriptions of classification, can be particularly useful in teaching classification in academic writing.

Trimble (1985) categorizes classification into three types: complete, partial, and implicit classification. He states that a complete classification in the form of 'X can be classified into … according to … ' should be taught to students in the first instance. However, I would argue that teaching classification this way may not be the best option. As shown in Table 31, complete classifications realized by Patterns 6, 13, and 21 occur only 18, 7, and 7 times in the corpus, respectively. The low frequency is likely due to the complexity of these three patterns which are not only the longest ones but are also in the passive voice. Starting with such complicated patterns is very likely to increase students' cognitive burden, potentially hindering their ability to flexibly organize the meaning components. Instead, I would recommend another teaching procedure: to scaffold the learning process by gradually increasing the complexity of patterns as students' proficiency develops. The teaching can start with simple and short patterns like 'Subclass + Hinge + Superclass'. Then one more meaning unit ('Indicator') can be added to the pattern to produce 'Subclass + Hinge + Indicator + Superclass'. Progressively, more complex patterns, such as those involving 'Basis' (e.g., Patterns 6, 13, and 21) and 'Categorizer' (e.g., Patterns 7, 15, and 16), can be introduced. This incremental procedure allows students to build confidence with simpler patterns before tackling more challenging ones, thereby reducing cognitive overload and enhancing comprehension.

The foregoing pedagogical implications can be concretized into a five-part course design to teach classification.

Part 1: Introduction. This part introduces students to the definition and function of classification in academic writing. Students could be told, for example, that classification is an important discourse act in academic writing that emphasizes the inclusion relation between entities, properties, or actions. The function of this discourse act is to reduce the complexity of abstract terminologies by organizing them into clearly identifiable groups.

Part 2: Passage reading and local grammar analyses. In Part 2, students are invited to read passages of classification. They may first read the ones filled with instances representing simple and short local grammar patterns, such as the sample passage presented below. The underlined parts are written based on the shortest local grammar patterns of classification, i.e., Patterns 1, 3, 9, and 10, each consisting of only three meaning components. More complex local grammar patterns can be introduced with longer passages at a later stage. After students finish reading each passage, the instructor can guide students to analyze the classification instances in each passage into local grammar patterns.

Sample passage. Solar power belongs to renewable energy (Pattern 1), which plays a pivotal role in sustainable development. The benefits of solar energy include reduced carbon emissions, lower energy costs, and minimal environmental impact (Pattern 9). Solar panel configurations are crucial in energy capture and storage. Common solar installation types are 1) rooftop systems, 2) ground-mounted arrays, and 3) floating solar farms (Pattern 10). Household solar equipment is usually the rooftop type (Pattern 3).

Part 3: Lexical resources. Part 3 provides lexical resources to facilitate the learning of certain terminologies with routinized lexical realizations. For example, the terminology 'Classifying' is often realized by verbs such as *include*, *classify*, *categoris(z)e*, *contain*, and *organis(z)e*. Moreover, 'Indicator' often contains classificatory nouns such as *type, category*, *kind*, *branch*, and *sort*. See Table 4 for a complete list.

Part 4: Contexts of use. Part 4 teaches students how to use the local grammar patterns in different contexts. This involves teaching

students how to interpret the context and identify the appropriate place where a certain pattern might fit. For example, there often comes a communicative situation where the reason for a classification requires justification. Students first need to recognize this communicative need and then realize the need by making sentences based on proper local grammar patterns containing 'Basis', such as 'Subclass + Hinge + Classifying + Superclass + Basis'.

Part 5: Relating to other discourse acts. By using local grammar patterns of classification as a starting point, the instructor can guide students to learn local grammars of other important academic discourse acts, such as definition and exemplification. Establishing this connection can help students identify the differences and similarities in the way meaning is expressed across different discourse acts. For instance, the classification pattern 'Superclass + Hinge + Subclass' can be compared with the definition pattern 'Definiendum + Hinge + Definiens' (Zhang & Su, 2022), while the classification pattern 'Subclass + Hinge + Indicator + Superclass' can facilitate the understanding of the exemplification pattern 'Exemplification + Hinge + Indicator + Exemplified' (Su & Zhang, 2020).

8. Conclusion

In this study, I have applied local grammar to the discourse act of classification in academic discourse and further revealed its implications for EAP writing pedagogy. Local grammar has transparently and context-specifically described the lexical-semantic regularities of classification, demonstrating how semantic roles can be mapped onto the corresponding linguistic forms of classification. In particular, the analyses show that nine semantic elements (e.g., 'Subclass', 'Superclass', 'Classifying', 'Indicator') are involved in performing classification in the DEAP Baby Corpus, resulting in 24 local grammar patterns. These patterns are embedded in different contexts and satisfy different communicative purposes. It has been found that local grammar can precisely quantify the lexical-semantic realizations of discourse acts in a manageable way. This study can also shed new light on terminology identification for local grammar studies on discourse acts that involve complex semantic relations. Furthermore, I have suggested a local-grammar-based pedagogy of classification, with increasingly complex patterns introduced as students' proficiency grows. Specifically, a five-part course design has been provided to clarify the teaching of classification in academic writing.

The study shows the benefits of using local grammar for analyzing a typical discourse act in academic discourse. Future studies in EAP settings can employ local grammar to analyze a wider range of discourse acts. Additionally, researchers can further conduct comparative studies to explore the differences and similarities in how classification is realized across various disciplines, between novice and expert writers, and in different sections of research articles. These investigations can deepen the understanding of classification in academic contexts and further contribute to EAP teaching practices.

Funding

This study is funded by the Social Science Planning Project of Anhui Province (grant number: AHSKQ2021D179).

References

Allen, C. M. (2005). *A local grammar of cause and effect: A corpus-driven study*. UK: University of Birmingham. Unpublished PhD dissertation.
Bai, Y., Zhang, R., Kong, F., Chen, J., & Mao, Y. (2021). Hypernym discovery via a recurrent mapping model. In C. Q. Zong, F. Xia, W. J. Li, & R. Navigli (Eds.), *Findings of the association for computational linguistics: ACL-IJCNLP 2021* (pp. 2912–2921). Stroudsburg: Association for Computational Linguistics.
Barnbrook, G. (2002). *Defining Language: A local grammar of definition sentences*. Amsterdam: John Benjamins Publishers.
Bielinskiene, A., Boizou, L., Kovalevskaite, J., & Utka, A. (2012). Towards the automatic extraction of term-defining contexts in Lithuanian. In A. Tavast, K. Muischnek, & M. Koit (Eds.), *Human Language technologies: The baltic perspective* (pp. 18–26). Amsterdam/Berlin/Tokyo/Washington DC: IOS Press.
Coffin, C., Curry, M. J., Goodman, S., Hewings, A., Lillis, T., & Swann, J. (2005). *Teaching academic writing: A toolkit for higher education*. London: Routledge.
Cruse, D. A. (2000). *Meaning in language*. Oxford: Oxford University Press.
Darian, S. (1997). *The language of classifying in introductory science texts. Journal of pragmatics, 27*(6), 815–839.
Faber, P. (2022). Frame-based terminology. In P. Faber, & M. C. L'Homme (Eds.), *Theoretical perspectives on terminology: Explaining terms, concepts and specialized knowledge, terminology and lexicography research and practice* (pp. 353–376). Amsterdam/Philadelphia: John Benjamins.
Faber, P., & López Rodríguez, C. I. (2012). Terminology and specialized language. In P. Faber (Ed.), *A cognitive linguistics view of terminology and specialized language* (pp. 9–32). Berlin/Boston: De Gruyter Mouton.
Faber, P., Verdejo-Román, J., León-Araúz, P., Reimerink, A., & Guzmán Pérez-Carrillo, G. (2014). Neural substrates of specialized knowledge representation: An fMRI study. *Revue Française de Linguistique Appliquée, 19*(1), 15–32.
Fang, Z. (2021). *Demystifying academic writing: Genres, moves, skills, and strategies*. London: Routledge.
Flachmann, K., Johnson, N., & Campbell, E. (2007). *Mosaics: Focusing on essays* (4th ed.). New Jersey: Pearson Prentice Hall.
Flowerdew, J. (2017). Corpus-based approaches to language description for specialized academic writing. *Language Teaching, 50*(1), 90–106.
Francis, G. (1993). A corpus-driven approach to grammar—principles, methods and examples. In M. Baker, G. Francis, & E. Tognini-Bonelli (Eds.), *Text and technology: In honour of john Sinclair* (pp. 137–156). Amsterdam: John Benjamins.
Francis, G., Hunston, S., & Manning, E. (1996). *Collins cobuild grammar patterns 1: Verbs*. New York: Harper Collins.
Gil Berrozpe, J. C. (2023). *Description, categorization, and Representation of Hyponymy in environmental terminology*. Spain: University of Granada. Unpublished PhD dissertation.
Hamp-Lyons, L., & Heasley, B. (2006). *Study writing: A course in written English for academic purposes*. Cambridge: Cambridge University Press.
Hearst, M. A. (1992). Automatic acquisition of hyponyms from large text corpora. In A. Zampolli (Ed.), *Proceedings of COLING-92* (pp. 539–545). Morristown (NJ): International Committee on Computational Linguistics.
Hearst, M. A. (1998). Automated discovery of word-net relations. In C. Fellbaum (Ed.), *WordNet: An electronic lexical Database* (pp. 132–152). Cambridge(MA)/London: MIT Press.
Hunston, S. (2002). Pattern grammar, language teaching, and linguistic variation: Applications of a corpus-driven grammar. In R. Reppen, S. Fitzmaurice, & D. Biber (Eds.), *Using corpora to explore linguistic variation* (pp. 167–183). Amsterdam: John Benjamins.
Hunston, S. (2015). Lexical grammar. In D. Biber, & R. Reppen (Eds.), *Cambridge handbook of corpus linguistics* (pp. 201–215). Cambridge: Cambridge University Press.
Hunston, S., & Francis, G. (2000). *Pattern grammar: A corpus-driven approach to the lexical grammar of English*. Amsterdam: John Benjamins.
Hunston, S., & Sinclair, J. (2000). A local grammar of evaluation. In S. Hunston, & G. Thompson (Eds.), *Evaluation in text: Authorial stance and the construction of discourse* (pp. 74–101). New York: Oxford University Press.

Hunston, S., & Su, H. (2019). Patterns, constructions, and local grammar: A case study of 'evaluation'. *Applied Linguistics, 40*(4), 567–593.
Kiniry, M., & Rose, M. (1993). *Critical strategies for academic thinking and writing: A text and reader* (2nd ed.). Boston: Bedford Books/St. Martin's Press.
Lakoff, G. (1987). *Women, fire and dangerous things: What categories reveal about the mind*. Chicago: University of Chicago Press.
León-Araúz, P., San Martín, A., & Faber, P. (2016). Pattern-based word sketches forthe extraction of semantic relations. In P. Drouin, N. Grabar, T. Hamon, K. Kageura, & K. Takeuchi (Eds.), *Proceedings of the 5th international workshop on computational terminology* (pp. 73–82). Osaka, Japan: The COLING 2016 Organizing Committee.
León-Araúz, P., San Martín, A., & Reimerink, A. (2018). The EcoLexicon English corpus as an open corpus in sketch engine. In J. Čibej, V. Gorjanc, I. Kosem, & S. Krek (Eds.), *Proceedings of the 18th EURALEX international congress* (pp. 893–901). Ljubljana: EURALEX.
Lewis, M. (2019). Compositional hyponymy with positive operators. In R. Mitkov, G. Angelova, & K. Bontcheva (Eds.), *Proceedings of recent advances in natural language processing* (pp. 638–647). Varna, Bulgaria: INCOMA Ltd.
Madsen, B. N., & Thomsen, H. E. (2009). Terminological concept modelling and conceptual data modelling. *International Journal of Metadata Semantics and Ontologies, 4*(4), 239–249.
Meteyard, L., Cuadrado, S. R., Bahrami, B., & Vigliocco, G. (2012). Coming of age: A review of embodiment and the neuroscience of semantics. *Cortex, 48*, 788–804.
Morley, J. (2023). *Academic Phrasebank*. Manchester: University of Manchester.
Murphy, M. L., & Koskela, A. (2010). *Key terms in semantics* (London/New York: Continuum).
Murray, N. (2012). *Writing essays in English language and linguistics: Principles, tips and strategies for undergraduates*. Cambridge: Cambridge University Press.
Murray, N., & Hughes, G. (2008). *Writing up your university assignments and research projects: A practical handbook*. London: McGraw-Hill. Education (UK).
Nirenburg, S., & Raskin, V. (2004). *Ontological semantics*. Cambridge (MA)/London: MIT Press.
Pearson, J. (1998). *Terms in context*. Amsterdam: Johns Benjamins.
Pontrandolfo, G., & Goźdź-Roszkowski, S. (2014). Exploring the local grammar of evaluation: The case of adjectival patterns in American and Italian judicial discourse. *Research in Language, 12*(1), 71–91.
Rose, M. (2013). Reclaiming the classroom. In G. DeLuca, L. Fox, M. A. Johnson, & M. Kogen (Eds.), *Dialogue on writing: Rethinking ESL, basic writing, and first-year composition* (pp. 9–27). London: Routledge.
Su, H. (2017). Local grammars of speech acts: An exploratory study. *Journal of Pragmatics, 111*, 72–83.
Su, H. (2020). Patterns, local grammars, and the design of English teaching materials. *ELT Journal, 74*(1), 73–82.
Su, H., & Lu, X. (2022). Assessing pragmatic performance in advanced L2 academic writing through the lens of local grammars: A case study of 'exemplification'. *Assessing Writing, 54*, 1–12.
Su, H., & Wei, N. (2018). "I'm really sorry about what I said" A local grammar of apology. *Pragmatics, 28*(3), 439–462.
Su, H., & Zhang, L. (2020). Local grammars and discourse acts in academic writing: A case study of exemplification in linguistics research articles. *Journal of English for Academic Purposes, 43*, 1–11.
Su, H., Zhang, Y., & Chau, M. H. (2022). Exemplification in Chinese English-major ma students' and expert writers' academic writing: A local grammar based investigation. *Journal of English for Academic Purposes, 58*, Article 101120.
Su, H., Zhang, Y., & Lu, X. (2021). Applying local grammars to the diachronic investigation of discourse acts in academic writing: The case of exemplification in Linguistics research articles. *English for Specific Purposes, 63*, 120–133.
Taylor, J. R. (1995). *Linguistic categorization: Prototypes in linguistic theory*. Oxford: Oxford University Press.
Trimble, L. (1985). *English for science and technology: A discourse approach*. Cambridge: Cambridge University Press.
Widdowson, H. G., & Allen, J. P. B. (1985). Teaching the communicative use of English. In J. Swales (Ed.), *Episodes in esp: A source and reference book on the development of English for science and technology* (pp. 69–89). Oxford: Pergamon Press.
Xu, J. J., & Sun, M. C. (2021). *The DEAP (Database of English for academic purposes) Baby corpus*. Beijing: Beijing Foreign Studies University.
Zhang, L., Jiang, R., & Zhang, J. (2024). 'Table 1 shows that…': A local grammar of graphic data commentary in discourse of economics. *English for Specific Purposes, 74*, 68–81.
Zhang, Y., & Su, H. (2022). "We define X as…": A local grammar of definition in linguistics research articles and its pedagogical value. *Journal of English for Academic Purposes, 59*, 1–12.

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